Q1: From the Groq and Openairouter, List down the Multimodels

Answer:

Multi models from https://developers.openai.com/api/docs/models/all

1. GPT-4.1: Input-Text, Image | Output-Text
2. GPT-5.4 pro: Input-Text, Image | Output-Text
3. GPT-5-Codex: Input-Text, Image | Output-Text
4. o4-mini-deep-research: Input-Text, Image | Output-Text
5. o3-deep-research: Input-Text, Image | Output-Text

Multi models from https://console.groq.com/home

1. Whisper | Audio - text
2. Canopy Labs Orpheus Arabic Saudi | Text – audio

Q2: Take 3 models from each service provider - https://openrouter.ai/models and list down context window, input output and token cost:

Answer:

1. xAI: Grok 4.20 Multi-Agent Beta: 2,000,000 | $2 | $6
2. OpenAI: GPT-5.4 Pro | 1,050,000 | $30 | $180
3. Xiaomi: MiMo-V2-Pro | 1,048,576 | $1 |$3

Q3: use the Tokenizer with different prompt and see how the tokens are generated

Answer:

GPT-5.x & O1/3 | GPT-4 & GPT-3.5 (legacy) | GPT-3 (legacy)

Artificial intelligence is an emerging technology in 2026 & also for the future.

Clear | Show example

Tokens	Characters
16	80

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Text | Token IDs

A helpful rule of thumb is that one token generally corresponds to ~4 characters of text for common English text. This translates to roughly ¾ of a word (so 100 tokens ~= 75 words).

If you need a programmatic interface for tokenizing text, check out our tiktoken package for Python. For JavaScript, the community-supported @dbdq/tiktoken package works with most GPT models.

16 80

[186671, 22990, 382, 448, 32073, 6340, 306, 220, 1323, 21, 827, 1217, 395, 290, 5277, 13]

Text | Token IDs

GPT-5.x & O1/3 | **GPT-4 & GPT-3.5 (legacy)** | GPT-3 (legacy)

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Tokens 17

Characters 80

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GPT-5.x & O1/3 | **GPT-4 & GPT-3.5 (legacy)** | GPT-3 (legacy)

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Clear | Show example

Tokens 17

Characters 80

```
[9470, 16895, 11478, 374, 459, 24084, 5557, 304, 220, 2366, 21, 612, 1101, 369,
279, 3938, 13]
```

Text | **Token IDs**

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GPT-5.x & O1/3 | GPT-4 & GPT-3.5 (legacy) | **GPT-3 (legacy)**

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GPT-5.x & O1/3 GPT-4 & GPT-3.5 (legacy) **GPT-3 (legacy)**

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Clear **Show example**

Tokens	Characters
16	80

```
[8001, 9542, 4430, 318, 281, 11823, 3037, 287, 1160, 2075, 1222, 635, 329, 262,
2003, 13]
```

Text **Token IDs**

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